

Hornet in Hollow Knight Game Plan

Goal

I want to be able to have Hornet from silksong be a toggleable character (press f5 or some other key and you switch between the knight and hornet like in the “knight in silksong” mod i really like that automatically keeps you as the original character and you can swap as needed in case of special animations to end a fight). Hornet will have all her movement abilities and combat abilities. This will include her dash, double jump, sprint, all her crests, all her tools, all of her special UI, getting silk from enemies on hit.

Stretch Goals

If I manage to incorporate these core features, I would also like to add:

If you play the needlin on enemies, the enemies in the radius will freeze (no custom freezing animation) and a random one will play a dreamnail dialogue.

Salubra's blessing is replaced with the Eva silk regeneration thing when sitting on benches.

Primal aspid projectiles inflict the maggot water effect

Plan for Implementation

Potential Phase 0 - Data Structures - maybe do, maybe do not do

I have already created a github where I have tried my darndest to attempt to port over some simple animations. Go through the 4/7/26 issues page to see what went wrong there. There aren't any changes to the actual functionality of the knight yet, so those are just physical changes, lucky for me.

This phase would encompass making sure that all the underlying systems: crests, tools, combat, unique animation “triggers” like the backflip for dashing, jumping, and going backwards I think, the specific steps might be different. There's probably some other unique things I might miss that i would rather just pull through all at once in the code.

Sprite handling systems should also be created. To my understanding, Hollow Knight uses tk2d, which not how silksong does sprite handling. That honestly sucks since now I have to figure out how to get that to work as well.

Possible Tips to base off of

. The "Architecture" Problem (HK vs Silksong)

HK is "simpler," but that's actually an advantage. No need to port the whole Silksong engine. Just need to **emulate** it within HK's hooks.

- **The "Silk" System:** In HK, this is just a variable. The Knight uses `PlayerData.instance.MPCharge`. To make Silk, you just create a new variable `float silkAmount`.
- **The Combat/Tools:** Don't try to make HK understand "Crests." Instead, write a `CrestManager` class in your mod that intercepts the `Attack` button. If `currentCrest == Wanderer`, run a specific function. HK doesn't need to "know" what a crest is; your mod just needs to tell HK what to do when a button is pressed.

2. Fixing the "Pivot & Atlas" Hell

The reason Claude's fix moved things up is likely because it was adjusting the **Transform position** instead of the **tk2d Sprite Anchor**.

- **The tk2d Reality:** HK uses `tk2dSprite`. Every sprite has an internal anchor point. If you replace the Knight's sprite with Hornet's, the game still thinks it's the Knight's 64×64 box.
- **The Fix:** You need to use `tk2dSprite.SetSprite(id)` and then manually offset the `transform.localPosition` based on the animation name.
 - *Example:* If `animation == "HornetDash"`, `transform.localPosition = new Vector3(0, -0.5f, 0)`.
- **Recommendation:** Use **UnityExplorer** while the game is running. Select the Hero object, find the Sprite component, and nudge the offsets until she sits on the floor. Copy those numbers into your code.

3. Implementing the Movement (The Hornet Way)

Since you can see the Silksong code, look for `HornetController`.

- **Don't copy the whole file.** Look specifically for the `FixedUpdate` logic for her **Sprint** and **Airdash**.
- **The Backflip:** This is a "State." In `HeroController`, the Knight has `HeroStates`. You will need to inject a new state using a **Finite State Machine (FSM)**. Most HK modders use `PlayMaker` hooks or simple C# timers to handle this.

4. Tactical "Phase 1" Plan

Forget the stretch goals (Needolin/Aspids) for now. Focus on the **Swap**:

1. Toggle Character (F5):

- Create a `MonoBehaviour` that listens for `Input.GetKeyDown(KeyCode.F5)`.
- When pressed, it toggles a `bool isHornet`.

2. The Hook:

- Hook into `On.HeroController.CanJump`. If `isHornet` is true, return your own custom `HornetCanJump()` logic.
- Hook into `On.HeroController.HeroDash`. Replace the dash speed/duration with Hornet's values from the Silksong DLL.

3. The Visuals:

- Instead of fighting the Atlas, look at the **"Custom Knight"** mod source code. It is the gold standard for how to swap sprites in HK without breaking pivots.

Shared Movement Animations

First, hornet should be properly implemented to do movements that both the knight and hornet share. This can be walking, jumping, double jumping, and the regular dash. Also includes resting/getting off benches, opening the map, taking damage. Any cases where hornet's animation doesn't work should also be set by default to revert to showing the knight's animation in case it can't find hornet's animation so as to not throw an error.

Basic nail strikes and the great nail slash (charge attack where you just hold down attack) are also ideally included, but depends on how hard it is to do crests.

Locking out Knight-Only Movement

The knight in hollow knight has special movement techniques that hornet can't use. These should be locked out.

This includes:

Crystal Dash, Shadow dash (dashes turn to normal dash), dreamnail.

In combat terms, this also includes the dash slash (charge attack + dash) and the spinning slash (charge attack + up/down keys)

New Movement

Next, we include the new movement options hornet can do. This includes:

Clawline, Sprint, Silk Soar.

All the unique animations for tools and crests should be set up here, to be added later.

UI

The hornet UI for health, silk, etc. should be there. In addition, the charms system should be replaced by the tools and crests.

Combat

A lot of smaller movement animations such as backflips and unique interactions with all of hornet's little animations should be added in. the tools systems should be 1000000% confirmed to work perfectly, same with crests.

Miscellaneous Functions

Togglable function to add in the silk maggot effect (drains silk over time status) when hit by a primal aspid.

Also bug fixes

Issues

4/8/26 request for help:

 I tried to make a Hornet in Hallownest mod

I would really appreciate some help here, I have no idea what I'm doing and all my attempts at fixing it seem to be worthless. DMing is fine, of course. i've watched though the guides and I feel either i don't get what to do or i am misunderstanding. the way i'm thinking of it is that i should find the extracted silksong scripts where they are coding about how to handle hornet's movement and animations and then do this equivalently in hollow knight, but i can't seem to figure it out. It doesn't help that the sprite sheets and atlases are so hard to understand.

Architectural Limitations: Porting "Backwards"

The primary hurdle identified is that Hollow Knight (HK) is not built to support the complexity of Hornet's kit. While "Knight in Silksong" mods are feasible because Silksong's engine is an evolution of HK's (retaining similar scripts with added fields), porting Hornet back into HK

requires the game to understand mechanics it simply wasn't designed for. Even still, I do want to throw my hat in the ring.

Complexity Gap: The Knight is mechanically simple; Hornet is highly complex. HK's engine lacks the "fields" and script parameters that define Hornet's unique physics and tool interactions. I don't know how this could be fixed. If i have to manually port over the silksong engine to hollow knight and then edit it to once again be hollow knight but basically revamped I'd be... very depressed. Add onto this the new update for hollow knight and I'd need to wait for the modding API to all be updated, and then also i'd like to wait for some mods I like to be updated and then I need to get the game on GoG again or ask someone to gift it since I lost access to the place i initially got it a long while back and then I'm very worried.

Enemy Compatibility: Every enemy in HK would potentially need an update to interact correctly with Hornet's tools and advanced AI/pathfinding requirements. Ideally this would only have to alter the code of hornet's tools that seek out enemies to understand hollow knight enemies as the same as silksong enemies. The pathfinding of enemies to hornet should be fine ideally since we'd be using the same pathfinding as the knight.

Alternative Approach: Porting Hallownest to Silksong

Due to the difficulty of modding HK, the discussion pivoted toward porting the Hallownest map into Silksong.

Optimization: Silksong is more optimized and already handles Hornet's physics natively.

Tool Dependency: This approach relies on upcoming, work-in-progress library mods:

AssetProxy: A WIP tool to help load HK assets into Silksong.

World Weaver: A custom scene development tool currently in the works.

Verdict: While "more stable" in theory, this would require waiting for these dev tools to release and would still involve massive script updates. This will only be done in the WORST CASE scenario where it is determined that there is no possible way to manage adding in hornet to the hollow knight game as a mod.